ML Engineer — Complete Roadmap

From Zero → Job-Ready · Every topic, subtopic, algorithm & what to search · 2025–26 Industry Standard

■ To Learn ✓ Already Done (green) → Algorithms inside ■ Exact search term

PHASE 1 Python, Math & Tools

Weeks 1–4 · The non-negotiable foundation

■ Python Core

TOPIC	SUBTOPICS / WHAT TO LEARN	ALGORITHMS / CONCEPTS	SEARCH THIS ON YT / DOCS
✓ **1. Python Basics**	· Variables, types (int, float, str, bool) · Type casting, input/output · String methods — split, join, strip, format · f-strings		■ ***Search:*** *Python basics for beginners full course*
✓ **2. Control Flow**	· if / elif / else · for loops, while loops · break, continue, pass · Nested loops		■ ***Search:*** *Python control flow tutorial*
✓ **3. Data Structures**	· Lists — append, pop, slice, sort · Tuples — immutable, unpacking · Dictionaries — keys, values, items · Sets — union, intersection, difference · List/dict/set comprehensions		■ ***Search:*** *Python data structures list dict set*
■ **4. Functions & Scope**	· def, return, default args · args, *kwargs · Lambda functions · Local vs global scope · Closures		■ ***Search:*** *Python functions advanced tutorial*
■ **5. OOP — Classes**	· class, __init__, self · Instance vs class attributes · Inheritance, super() · Dunder methods — __str__, __len__, __eq__ · Encapsulation, polymorphism		■ ***Search:*** *Python OOP object oriented programming*
■ **6. Advanced Python**	· Generators & yield · Decorators (@) · Context managers (with) · Iterators & itertools · Map, filter, zip		■ ***Search:*** *Python generators decorators advanced*
✓ **7. File I/O & Error Handling**	· Open, read, write files · CSV, JSON read/write · try / except / finally · Custom exceptions · os, sys, pathlib modules		■ ***Search:*** *Python file handling exception handling*
✓ **8. Modules & Environment**	· import, from, as · pip, requirements.txt · Virtual environments (venv) · __name__ == '__main__' · Package structure		■ ***Search:*** *Python virtual environment pip packages*

■ NumPy & Pandas

TOPIC	SUBTOPICS / WHAT TO LEARN	ALGORITHMS / CONCEPTS	SEARCH THIS ON YT / DOCS
■ **9. NumPy**	· np.array, shape, dtype, reshape · Indexing, slicing, boolean mask · Broadcasting rules · np.zeros, ones, eye, random · Aggregations — sum, mean, std, max · np.dot, np.linalg (norm, inv, eig) · Stack, concatenate, split	→ *Array operations* → *Broadcasting* → *Linear algebra ops*	■ ***Search:*** *NumPy complete tutorial Python*
■ **10. Pandas**	· Series and DataFrame creation · read_csv, read_json, read_excel · head, tail, info, describe · Indexing — loc, iloc, at · Filtering — boolean indexing · groupby, agg, transform · merge, join, concat · fillna, dropna, isnull · apply, map, applymap · pivot_table, crosstab	→ *DataFrame ops* → *Group aggregation* → *Merge strategies*	■ ***Search:*** *Pandas complete tutorial data analysis*
■ **11. Matplotlib & Seaborn**	· Line, bar, scatter, histogram plots · Subplots, figure size, labels · Seaborn — heatmap, pairplot, boxplot · Distribution plots, violin plot · Customization — colors, legends, grids		■ ***Search:*** *Matplotlib Seaborn data visualization Python*

■ Mathematics for ML

TOPIC	SUBTOPICS / WHAT TO LEARN	ALGORITHMS / CONCEPTS	SEARCH THIS ON YT / DOCS
■ **12. Linear Algebra**	· Scalars, vectors, matrices, tensors · Matrix addition, scalar multiplication · Matrix multiplication (dot product) · Transpose, inverse, determinant · Identity matrix, diagonal matrix · Rank, null space, column space · Norm — L1, L2, Frobenius	→ *Dot product* → *Matrix inverse* → *Gram-Schmidt*	■ ***Search:*** *Linear algebra for machine learning 3Blue1Brown*
■ **13. Eigenvalues & SVD**	· Eigenvalues and eigenvectors · Eigendecomposition · Singular Value Decomposition (SVD) · Low-rank approximation · Applications in PCA	→ *SVD* → *Eigendecomposition* → *PCA math*	■ ***Search:*** *Eigenvalues eigenvectors SVD explained*
■ **14. Calculus**	· Derivatives — power, chain, product rule · Partial derivatives · Gradients and directional derivatives · Jacobian matrix · Hessian matrix · Integrals (basic understanding)	→ *Gradient* → *Chain rule* → *Jacobian*	■ ***Search:*** *Calculus for deep learning gradient descent*

■ **15. Probability**	· Sample space, events, probability · Conditional probability, Bayes theorem · Joint, marginal, conditional distributions · Independence, mutual exclusivity · Random variables — discrete, continuous · PMF, PDF, CDF · Expected value, variance	→ *Bayes theorem* → *Probability distributions* → *Conditional probability*	■ ***Search:*** *Probability for machine learning statistics*
■ **16. Probability Distributions**	· Bernoulli, Binomial distribution · Normal (Gaussian) distribution · Poisson distribution · Uniform distribution · Exponential distribution · Multivariate Gaussian · Central Limit Theorem	→ *Gaussian distribution* → *Binomial* → *CLT*	■ ***Search:*** *Probability distributions machine learning*
■ **17. Statistics**	· Mean, median, mode · Variance, std deviation, IQR · Skewness, kurtosis · Covariance and correlation · Pearson, Spearman correlation · Hypothesis testing — null vs alternative · p-value, significance level · t-test, chi-squared test, ANOVA · Confidence intervals · Type I / Type II errors	→ *t-test* → *Chi-squared* → *ANOVA* → *p-value*	■ ***Search:*** *Statistics for data science hypothesis testing*
■ **18. Information Theory**	· Entropy — Shannon entropy · Cross-entropy loss · KL Divergence · Mutual information · Information gain (used in Decision Trees)	→ *Shannon entropy* → *KL Divergence* → *Cross-entropy*	■ ***Search:*** *Information theory machine learning entropy*

■ Tools & Environment

TOPIC	SUBTOPICS / WHAT TO LEARN	ALGORITHMS / CONCEPTS	SEARCH THIS ON YT / DOCS
✓ **19. Git & GitHub**	· init, add, commit, push, pull · Branching — branch, checkout, merge · Rebase vs merge · Pull requests, code review · Resolving merge conflicts · .gitignore · Tags, releases		■ ***Search:*** *Git GitHub complete tutorial*
✓ **20. Jupyter Notebooks**	· Cells — code vs markdown · Keyboard shortcuts · Magic commands — %timeit, %%time · Widgets, interactive plots · Export to PDF/HTML		■ ***Search:*** *Jupyter notebook tips tricks tutorial*
✓ **21. VS Code for ML**	· Python extension, linting (flake8) · Debugger — breakpoints, watch · Jupyter inside VS Code · Remote SSH extension · GitHub Copilot basics		■ ***Search:*** *VS Code Python ML setup tutorial*

■ **22. Linux / Bash**	· ls, cd, mkdir, rm, cp, mv · grep, find, awk, sed · Pipes and redirection · chmod, chown · Cron jobs · Shell scripts (.sh) · SSH, scp		■ ***Search:*** *Linux command line for data science*

PHASE 2

Core Machine Learning

Weeks 5–10 · The bread and butter

■ ML Fundamentals

TOPIC	SUBTOPICS / WHAT TO LEARN	ALGORITHMS / CONCEPTS	SEARCH THIS ON YT / DOCS
■ 23. ML Concepts	· Supervised, unsupervised, semi-supervised · Reinforcement learning overview · Online vs batch learning · Parametric vs non-parametric models · Discriminative vs generative models · The ML workflow end to end		■ ***Search:*** *Machine learning types supervised unsupervised*
■ 24. Data Splitting & Leakage	· Train / validation / test split · Stratified split · Time-series split · Data leakage — what it is, how to avoid · Hold-out vs cross-validation		■ ***Search:*** *train test split data leakage machine learning*
■ 25. Bias-Variance Tradeoff	· Bias — underfitting · Variance — overfitting · Bias-variance decomposition · Learning curves · Regularization as variance reducer · Model complexity vs error	→ *Bias-variance tradeoff* → *Regularization*	■ ***Search:*** *bias variance tradeoff explained machine learning*
■ 26. Feature Engineering	· One-hot encoding, label encoding, ordinal · Min-max scaling, standardization (z-score) · Robust scaler, power transform · Binning / discretization · Polynomial features · Interaction features · Target encoding · Date/time feature extraction · Text to features — TF-IDF, BoW	→ *StandardScaler* → *LabelEncoder* → *PolynomialFeatures* → *TF-IDF*	■ ***Search:*** *feature engineering machine learning tutorial*
■ 27. Feature Selection	· Filter methods — correlation, chi-squared · Wrapper — RFE (Recursive Feature Elimination) · Embedded — Lasso, Ridge coefficients · Variance inflation factor (VIF) · Permutation importance · SHAP values for feature importance	→ *RFE* → *Lasso selection* → *SHAP* → *VIF*	■ ***Search:*** *feature selection techniques machine learning*
■ 28. Handling Missing Data	· Types — MCAR, MAR, MNAR · Dropping rows/cols (when safe) · Mean/median/mode imputation · KNN imputation · Iterative imputation (MICE) · Indicator variables for missingness	→ *SimpleImputer* → *KNNImputer* → *IterativeImputer*	■ ***Search:*** *handling missing data machine learning imputation*

■ **29. Imbalanced Data**	· Class imbalance problem · Oversampling — SMOTE, ADASYN · Undersampling — RandomUnderSampler · Hybrid methods · class_weight parameter · Threshold tuning · Evaluation with imbalanced data	→ *SMOTE* → *ADASYN* → *class_weight*	■ ***Search:*** *imbalanced dataset machine learning SMOTE*

■ Supervised Learning — Regression

TOPIC	SUBTOPICS / WHAT TO LEARN	ALGORITHMS / CONCEPTS	SEARCH THIS ON YT / DOCS
■ **30. Linear Regression**	· Simple and multiple linear regression · OLS — Ordinary Least Squares · Gradient descent from scratch · Assumptions — linearity, homoscedasticity, normality · Residual analysis · Adjusted R^2, AIC, BIC · Multicollinearity	→ *OLS* → *Gradient Descent* → *Ridge/Lasso*	■ ***Search:*** *linear regression from scratch Python numpy*
■ **31. Regularization**	· Ridge Regression (L2) · Lasso Regression (L1) · Elastic Net (L1 + L2) · Alpha hyperparameter tuning · Feature selection via Lasso · Weight decay intuition	→ *Ridge* → *Lasso* → *ElasticNet* → *Alpha tuning*	■ ***Search:*** *Ridge Lasso Elastic Net regularization Python*
■ **32. Polynomial & Other Regression**	· Polynomial features + linear regression · Spline regression · Quantile regression · Poisson regression · Regression with interaction terms	→ *PolynomialFeatures* → *Splines*	■ ***Search:*** *polynomial regression Python scikit-learn*

■ Supervised Learning — Classification

TOPIC	SUBTOPICS / WHAT TO LEARN	ALGORITHMS / CONCEPTS	SEARCH THIS ON YT / DOCS
■ **33. Logistic Regression**	· Sigmoid function, log-odds · Binary and multiclass (OvR, OvO) · Decision boundary · Log loss (cross-entropy) · Softmax for multiclass · Regularization in logistic regression	→ *Sigmoid* → *Softmax* → *Log loss* → *OvR/OvO*	■ ***Search:*** *logistic regression explained from scratch*
■ **34. Naive Bayes**	· Bayes theorem application · Gaussian Naive Bayes · Multinomial Naive Bayes (text) · Bernoulli Naive Bayes · Laplace smoothing · When NB works well	→ *GaussianNB* → *MultinomialNB* → *Laplace smoothing*	■ ***Search:*** *Naive Bayes classifier Python machine learning*
■ **35. Decision Trees**	· Tree structure — nodes, branches, leaves · Splitting criteria — Gini, Entropy, Info Gain · Pruning — pre and post pruning · Max depth, min samples split · Feature importance from trees · Regression trees	→ *Gini impurity* → *Information Gain* → *Entropy* → *DecisionTreeClassifier*	■ ***Search:*** *Decision Tree algorithm explained Python*

■ 36. Random Forest	· Bagging — Bootstrap Aggregating · Feature randomness (mtry) · OOB (out of bag) error · Feature importance · Random forest for regression · Hyperparams — n_estimators, max_depth	*→ Bagging* *→ OOB error* *→ RandomForestClassifier*	**■ *Search:*** *Random Forest explained scikit-learn Python*
■ 37. Gradient Boosting	· Boosting concept — weak to strong learners · AdaBoost — adaptive boosting · Gradient Boosting Machines (GBM) · XGBoost — trees + regularization · LightGBM — leaf-wise growth · CatBoost — categorical handling · Key hyperparams — learning rate, n_estimators, subsample · Early stopping	*→ AdaBoost* *→ XGBoost* *→ LightGBM* *→ CatBoost* *→ Learning rate*	**■ *Search:*** *XGBoost LightGBM gradient boosting Python tutorial*
■ 38. SVM	· Hyperplane and margin · Support vectors · Hard vs soft margin (C parameter) · Kernel trick — linear, RBF, polynomial, sigmoid · SVM for regression (SVR) · Multiclass SVM	*→ Linear kernel* *→ RBF kernel* *→ C parameter* *→ SVR*	**■ *Search:*** *SVM Support Vector Machine kernel trick Python*
■ 39. K-Nearest Neighbors	· Distance metrics — Euclidean, Manhattan, Minkowski · Choosing K · Weighted KNN · KNN for regression · Curse of dimensionality · KD-Tree and Ball-Tree for fast search	*→ KNeighborsClassifier* *→ Euclidean distance* *→ KD-Tree*	**■ *Search:*** *KNN K Nearest Neighbors Python machine learning*

■ Unsupervised Learning

TOPIC	SUBTOPICS / WHAT TO LEARN	ALGORITHMS / CONCEPTS	SEARCH THIS ON YT / DOCS
■ 40. Clustering	· K-Means — algorithm, elbow method, inertia · K-Means++ initialization · DBSCAN — density-based, eps, min_samples · Hierarchical — agglomerative, dendrogram · Gaussian Mixture Models (GMM) · Silhouette score, Davies-Bouldin index	*→ K-Means* *→ DBSCAN* *→ GMM* *→ Silhouette score*	**■ *Search:*** *clustering algorithms machine learning K-Means DBSCAN*
■ 41. Dimensionality Reduction	· PCA — principal components, explained variance · t-SNE — non-linear, perplexity param · UMAP — faster, topology-preserving · LDA — Linear Discriminant Analysis · ICA — Independent Component Analysis · Autoencoders (intro)	*→ PCA* *→ t-SNE* *→ UMAP* *→ LDA*	**■ *Search:*** *dimensionality reduction PCA tSNE UMAP Python*

■ 42. Anomaly Detection	· Statistical methods — z-score, IQR · Isolation Forest · One-Class SVM · Local Outlier Factor (LOF) · Autoencoders for anomaly detection · Use cases — fraud, network intrusion	→ *IsolationForest* → *LOF* → *One-Class SVM*	■ ***Search:*** *anomaly detection machine learning Python*

■ Model Evaluation & Tuning

TOPIC	SUBTOPICS / WHAT TO LEARN	ALGORITHMS / CONCEPTS	SEARCH THIS ON YT / DOCS
■ 43. Classification Metrics	· Accuracy — when to use/avoid · Precision, Recall, F1 score · Macro, micro, weighted average · Confusion matrix analysis · ROC curve, AUC score · Precision-Recall curve · Log loss / binary cross-entropy · Matthews Correlation Coefficient · Cohen's Kappa	→ *classification_report* → *roc_auc_score* → *confusion_matrix* → *f1_score*	■ ***Search:*** *classification metrics precision recall F1 ROC AUC*
■ 44. Regression Metrics	· MAE — Mean Absolute Error · MSE — Mean Squared Error · RMSE — Root Mean Squared Error · R^2 — coefficient of determination · Adjusted R^2 · MAPE — Mean Absolute Percentage Error · Huber loss, quantile loss	→ *mean_absolute_error* → *r2_score* → *RMSE* → *MAPE*	■ ***Search:*** *regression metrics MAE RMSE R squared Python*
■ 45. Cross Validation	· K-fold cross validation · Stratified K-fold · Leave-One-Out (LOO) · Time-series split (no data leakage) · Repeated K-fold · Nested cross-validation	→ *KFold* → *StratifiedKFold* → *cross_val_score* → *TimeSeriesSplit*	■ ***Search:*** *cross validation K-fold Python scikit-learn*
■ 46. Hyperparameter Tuning	· GridSearchCV — exhaustive search · RandomizedSearchCV — random sampling · Bayesian optimization (Optuna, Hyperopt) · Early stopping · Param grids and distributions · Pipeline + GridSearch	→ *GridSearchCV* → *RandomizedSearchCV* → *Optuna* → *Hyperopt*	■ ***Search:*** *hyperparameter tuning Optuna GridSearch Python*
■ 47. Pipelines	· sklearn Pipeline object · ColumnTransformer · Custom transformers (BaseEstimator) · Feature union · Pipeline + cross-validation · Saving pipelines with joblib	→ *Pipeline* → *ColumnTransformer* → *FunctionTransformer*	■ ***Search:*** *scikit-learn Pipeline ColumnTransformer tutorial*

PHASE 3

Deep Learning & NLP

Weeks 11–18 · High-value skills

■ Neural Network Fundamentals

TOPIC	SUBTOPICS / WHAT TO LEARN	ALGORITHMS / CONCEPTS	SEARCH THIS ON YT / DOCS
■ **48. Neural Networks from Scratch**	· Perceptron — weights, bias, activation · Multi-layer perceptron (MLP) · Forward pass — matrix multiplication · Loss computation · Backpropagation — chain rule · Weight update — gradient descent · Implement in pure NumPy	→ *Perceptron* → *Forward pass* → *Backprop* → *MLP*	■ ***Search:*** *neural network from scratch numpy Python*
■ **49. Activation Functions**	· Sigmoid — vanishing gradient problem · Tanh — centered sigmoid · ReLU — dying ReLU problem · Leaky ReLU, ELU, SELU · GELU — used in transformers · Softmax — multiclass output · When to use which	→ *ReLU* → *GELU* → *Softmax* → *Leaky ReLU*	■ ***Search:*** *activation functions neural networks explained*
■ **50. Optimizers**	· SGD — stochastic gradient descent · Momentum — accelerate SGD · RMSprop — adaptive learning rate · Adam — momentum + RMSprop · AdamW — Adam + weight decay · Learning rate schedulers · Gradient clipping	→ *SGD* → *Adam* → *AdamW* → *RMSprop* → *LR scheduler*	■ ***Search:*** *optimizers machine learning Adam SGD explained*
■ **51. Regularization in DL**	· Dropout — randomly zero neurons · Batch Normalization · Layer Normalization · Weight decay (L2 in optimizer) · Early stopping · Data augmentation · Label smoothing	→ *Dropout* → *BatchNorm* → *LayerNorm* → *Early stopping*	■ ***Search:*** *regularization deep learning dropout batch norm*
■ **52. Training Deep Networks**	· Mini-batch gradient descent · Epoch, batch size, iterations · Weight initialization — Xavier, He · Vanishing & exploding gradients · Gradient clipping · Learning rate finding (lr_finder) · Training loop from scratch (PyTorch)	→ *Xavier init* → *He init* → *Gradient clipping* → *Mini-batch*	■ ***Search:*** *training deep neural network PyTorch from scratch*

■ PyTorch & Keras

TOPIC	SUBTOPICS / WHAT TO LEARN	ALGORITHMS / CONCEPTS	SEARCH THIS ON YT / DOCS
■ **53. PyTorch Core**	· Tensors — creation, ops, GPU · Autograd — automatic differentiation · nn.Module — build models · nn.Linear, nn.ReLU, nn.Dropout · DataLoader, Dataset class · Training loop — zero_grad, backward, step · Save/load model — state_dict · Move to GPU — .to(device)	→ *torch.Tensor* → *nn.Module* → *DataLoader* → *autograd*	■ ***Search:*** *PyTorch tutorial complete beginners*

■ **54. Keras / TensorFlow**	· Sequential and Functional API · Dense, Activation, Dropout layers · Compile — optimizer, loss, metrics · fit, evaluate, predict · Callbacks — EarlyStopping, ModelCheckpoint · Custom layers and training loops · TensorBoard for visualization	→ *Sequential* → *Functional API* → *Callbacks* → *TensorBoard*	■ ***Search:*** *Keras TensorFlow tutorial deep learning*

■ Convolutional Neural Networks

TOPIC	SUBTOPICS / WHAT TO LEARN	ALGORITHMS / CONCEPTS	SEARCH THIS ON YT / DOCS
■ **55. CNNs**	· Convolution operation — kernel, stride, padding · Pooling — max pooling, average pooling · Feature maps, receptive field · CNN architecture — conv-bn-relu-pool · Flattening + fully connected · Transfer learning — pretrained CNN · Data augmentation for images	→ *Conv2d* → *MaxPool2d* → *BatchNorm2d* → *Transfer learning*	■ ***Search:*** *CNN convolutional neural network explained PyTorch*
■ **56. CNN Architectures**	· LeNet-5 — original CNN · AlexNet — deep CNN, ReLU, dropout · VGG — very deep, small kernels · ResNet — skip connections, residuals · Inception / GoogLeNet · EfficientNet — compound scaling · MobileNet — lightweight CNNs	→ *ResNet* → *VGG* → *EfficientNet* → *MobileNet*	■ ***Search:*** *ResNet EfficientNet architecture explained*

■ Recurrent Networks & Sequence Models

TOPIC	SUBTOPICS / WHAT TO LEARN	ALGORITHMS / CONCEPTS	SEARCH THIS ON YT / DOCS
■ **57. RNNs & LSTMs**	· RNN — hidden state, vanishing gradient · LSTM — cell state, gates (forget, input, output) · GRU — gated recurrent unit · Bidirectional RNN/LSTM · Sequence-to-sequence models · Many-to-one, many-to-many · Teacher forcing	→ *LSTM* → *GRU* → *Bidirectional* → *Seq2seq*	■ ***Search:*** *LSTM RNN recurrent neural network PyTorch explained*
■ **58. Time Series with DL**	· Sliding window approach · Lag features · LSTM for time series forecasting · Temporal Convolutional Network (TCN) · Transformer for time series · Stationarity, differencing · ARIMA baseline	→ *LSTM forecasting* → *TCN* → *ARIMA* → *Stationarity*	■ ***Search:*** *time series forecasting LSTM Python tutorial*

■ NLP Fundamentals

TOPIC	SUBTOPICS / WHAT TO LEARN	ALGORITHMS / CONCEPTS	SEARCH THIS ON YT / DOCS

■ 59. Text Preprocessing	· Tokenization — word, sentence, subword · Lowercasing, punctuation removal · Stop word removal · Stemming (Porter, Snowball) · Lemmatization (WordNet, spaCy) · Regular expressions for text · Vocabulary building, OOV tokens	→ *nltk* → *spaCy* → *re module* → *Stemming* → *Lemmatization*	■ ***Search:*** *NLP text preprocessing Python NLTK spaCy*
✓ 60. Text Vectorization	· Bag of Words (BoW) · TF-IDF — term frequency-inverse document freq · N-grams (bigrams, trigrams) · CountVectorizer, TfidfVectorizer · Cosine similarity · BM25 ranking	→ *CountVectorizer* → *TfidfVectorizer* → *cosine_similarity* → *BM25*	■ ***Search:*** *TF-IDF bag of words text vectorization Python*
■ 61. Word Embeddings	· Word2Vec — CBOW and Skip-gram · GloVe — global vectors · FastText — subword embeddings · Embedding layer in PyTorch/Keras · Pre-trained embeddings loading · Sentence embeddings	→ *Word2Vec* → *GloVe* → *FastText* → *nn.Embedding*	■ ***Search:*** *Word2Vec GloVe word embeddings Python explained*
✓ 62. Classic NLP Tasks	· Text classification · Sentiment analysis · Named Entity Recognition (NER) · POS tagging · Dependency parsing · Text summarization · Machine translation basics	→ *spaCy NER* → *TextClassification* → *POS tagging*	■ ***Search:*** *NLP tasks text classification NER sentiment Python*

■ Transformers & Modern NLP

TOPIC	SUBTOPICS / WHAT TO LEARN	ALGORITHMS / CONCEPTS	SEARCH THIS ON YT / DOCS
■ 63. Attention Mechanism	· Seq2seq with attention (Bahdanau) · Self-attention · Scaled dot-product attention · Query, Key, Value (Q, K, V) · Multi-head attention · Attention masks and padding	→ *Self-attention* → *Q K V* → *Multi-head attention* → *Bahdanau*	■ ***Search:*** *attention mechanism transformer explained from scratch*
■ 64. Transformer Architecture	· Encoder-decoder structure · Positional encoding · Feed-forward sublayer · Layer normalization · Residual connections · Encoder-only (BERT) · Decoder-only (GPT) · Encoder-decoder (T5, BART)	→ *Positional encoding* → *BERT* → *GPT* → *T5* → *BART*	■ ***Search:*** *transformer architecture explained Attention is All You Need*
✓ 65. HuggingFace Transformers	· AutoTokenizer, AutoModel · pipeline() API · from_pretrained loading · Fine-tuning with Trainer API · Dataset and DataCollator · Evaluate metrics · Push to Hub	→ *AutoTokenizer* → *Trainer* → *pipeline* → *DataCollator*	■ ***Search:*** *HuggingFace transformers fine-tuning tutorial*

✓ **66. BERT & Variants**	· BERT — masked language model · RoBERTa — robust BERT · DistilBERT — smaller, faster · ALBERT — parameter sharing · DeBERTa — disentangled attention · Sentence-BERT (SBERT) · Fine-tuning BERT for classification	*→ BERT* *→ RoBERTa* *→ DistilBERT* *→ SBERT* *→ Fine-tuning*	■ ***Search:*** *BERT fine-tuning text classification HuggingFace*
✓ **67. KeyBERT & Keyword Extraction**	· KeyBERT — BERT-based keyword extraction · YAKE — unsupervised keyword extraction · RAKE algorithm · Topic modeling — LDA · BERTopic	*→ KeyBERT* *→ YAKE* *→ RAKE* *→ LDA* *→ BERTopic*	■ ***Search:*** *keyword extraction KeyBERT Python NLP*

PHASE 4

LLMs & Agentic AI

Weeks 19–24 · The most in-demand 2025–26 skill

■ LLM Fundamentals

TOPIC	SUBTOPICS / WHAT TO LEARN	ALGORITHMS / CONCEPTS	SEARCH THIS ON YT / DOCS
■ 68. How LLMs Work	· Tokenization — BPE, WordPiece, SentencePiece · Next token prediction · Temperature, top-k, top-p sampling · Context window, tokens · KV cache · Inference vs training · Model sizes — parameters, FLOPs	*→ BPE tokenizer* *→ Temperature sampling* *→ KV cache* *→ Top-p*	■ ***Search:*** *how large language models work explained*
✓ 69. Prompt Engineering	· Zero-shot prompting · Few-shot prompting · Chain-of-thought (CoT) · Tree-of-thought (ToT) · ReAct pattern · System prompts · Prompt injection and defense · Structured output prompting	*→ Zero-shot* *→ Few-shot* *→ Chain-of-thought* *→ ReAct* *→ ToT*	■ ***Search:*** *prompt engineering techniques LLM tutorial*
■ 70. Fine-tuning LLMs	· Full fine-tuning vs PEFT · LoRA — Low-Rank Adaptation · QLoRA — quantized LoRA · Adapter layers · Instruction fine-tuning · RLHF — Reinforcement Learning from Human Feedback · DPO — Direct Preference Optimization · Unsloth for fast fine-tuning	*→ LoRA* *→ QLoRA* *→ RLHF* *→ DPO* *→ PEFT library*	■ ***Search:*** *LoRA QLoRA fine-tuning LLM Llama tutorial*
✓ 71. RAG — Retrieval Augmented Generation	· RAG architecture — retriever + generator · Document chunking strategies · Embedding models for retrieval · Vector similarity — cosine, dot product · FAISS — Facebook AI similarity search · pgvector — PostgreSQL vector extension · Pinecone, Weaviate, Chroma · Re-ranking — cross-encoder · Naive RAG vs advanced RAG · Evaluation — RAGAS framework	*→ FAISS* *→ pgvector* *→ Chroma* *→ RAGAS* *→ Cross-encoder*	■ ***Search:*** *RAG retrieval augmented generation tutorial*
■ 72. LLM Evaluation	· BLEU, ROUGE scores · Perplexity · BERTScore · RAGAS — faithfulness, relevancy · LLM-as-judge · Hallucination detection · Human evaluation frameworks	*→ BLEU* *→ ROUGE* *→ BERTScore* *→ RAGAS* *→ Perplexity*	■ ***Search:*** *LLM evaluation metrics RAGAS hallucination*

■ Agentic AI & Frameworks

TOPIC	SUBTOPICS / WHAT TO LEARN	ALGORITHMS / CONCEPTS	SEARCH THIS ON YT / DOCS

✓ 73. LangChain	· LLMChain, PromptTemplate · Memory — ConversationBufferMemory · Retrievers and vector stores · Agents and tools · Sequential chains · Callbacks and streaming · LCEL — LangChain Expression Language	→ *LLMChain* → *ConversationBufferMemory* → *LCEL* → *AgentExecutor*	■ ***Search:*** *LangChain tutorial Python complete*
✓ 74. LangGraph	· Graph-based agent workflows · StateGraph, nodes, edges · Conditional edges · Checkpointing and persistence · Human-in-the-loop · Multi-agent graphs · Tool calling in graphs	→ *StateGraph* → *Conditional edges* → *Checkpointing* → *HITL*	■ ***Search:*** *LangGraph tutorial agentic AI Python*
✓ 75. CrewAI & Multi-Agent	· Agents — role, goal, backstory · Tasks — description, expected output · Crew — agent orchestration · Process — sequential, hierarchical · Tool use in CrewAI · Inter-agent communication · Memory types in CrewAI	→ *Agent* → *Task* → *Crew* → *Process* → *Tool*	■ ***Search:*** *CrewAI multi-agent framework tutorial*
✓ 76. AutoGen & BeeAI	· ConversableAgent in AutoGen · Human proxy agent · Group chat manager · Code execution agents · BeeAI — IBM framework basics · Agent memory and tools · Multi-agent debate pattern	→ *ConversableAgent* → *GroupChat* → *UserProxyAgent*	■ ***Search:*** *AutoGen multi-agent conversation Python*
■ 77. Tool Use & Function Calling	· OpenAI function calling · Anthropic tool use · JSON schema for tools · Parallel tool calling · Tool result handling · Structured outputs · MCP — Model Context Protocol	→ *function_calling* → *tool_use* → *JSON schema* → *MCP*	■ ***Search:*** *LLM function calling tool use API tutorial*
✓ 78. Vector Databases Deep Dive	· Embeddings — what they are · FAISS — IVF, HNSW indexes · pgvector — ivfflat, hnsw index · Pinecone — serverless vectors · Chroma — local vector DB · Weaviate — hybrid search · Qdrant — filtering + vectors · Hybrid search — BM25 + vector	→ *FAISS HNSW* → *pgvector hnsw* → *Hybrid search* → *Pinecone*	■ ***Search:*** *vector database FAISS pgvector Pinecone tutorial*

PHASE 5

MLOps & Deployment

Weeks 25–30 · Get models to production

■ Model Serving & APIs

TOPIC	SUBTOPICS / WHAT TO LEARN	ALGORITHMS / CONCEPTS	SEARCH THIS ON YT / DOCS
✓ 79. FastAPI for ML	· Route, request, response models · Pydantic validation · Loading model at startup (lifespan) · Async endpoints · Background tasks · File uploads · API authentication (JWT, API keys) · OpenAPI docs	→ FastAPI → Pydantic → lifespan → APIRouter	■ **Search:** *FastAPI machine learning model deployment tutorial*
■ 80. Model Serialization	· pickle — Python objects · joblib — sklearn models (faster) · ONNX — cross-platform model format · TorchScript — PyTorch to production · SavedModel — TensorFlow format · Model versioning best practices	→ pickle → joblib → ONNX → TorchScript	■ **Search:** *model serialization deployment pickle joblib ONNX*
✓ 81. Streamlit & Gradio	· Streamlit — layout, widgets, charts · Session state management · File upload and display · Caching with @st.cache_data · Gradio — quick ML demos · Sharing via Streamlit Cloud / HuggingFace Spaces	→ st.cache_data → gr.Interface → Streamlit Cloud	■ **Search:** *Streamlit machine learning app tutorial*

■ Docker & Containerization

TOPIC	SUBTOPICS / WHAT TO LEARN	ALGORITHMS / CONCEPTS	SEARCH THIS ON YT / DOCS
■ 82. Docker	· Dockerfile — FROM, RUN, COPY, CMD · docker build, run, ps, stop · Docker images and layers · docker-compose — multi-service apps · Volumes and bind mounts · Environment variables in Docker · Push to Docker Hub · Multi-stage builds	→ Dockerfile → docker-compose → docker build → Multi-stage	■ **Search:** *Docker tutorial machine learning deployment*
■ 83. Docker for ML	· Containerize FastAPI + ML model · GPU support — nvidia-docker · Dockerfile for PyTorch/TF · Managing model weights in containers · Health checks · Resource limits — CPU, memory	→ nvidia-docker → CUDA container → GPU Docker	■ **Search:** *Docker GPU machine learning PyTorch container*

■ Cloud & Infrastructure

TOPIC	SUBTOPICS / WHAT TO LEARN	ALGORITHMS / CONCEPTS	SEARCH THIS ON YT / DOCS
■ 84. Cloud Basics (AWS)	· S3 — object storage, boto3 · EC2 — virtual machines · Lambda — serverless functions · SageMaker — managed ML platform · ECR — container registry · IAM — roles and permissions · VPC basics	→ boto3 → S3 → EC2 → SageMaker → Lambda	■ **Search:** *AWS machine learning S3 SageMaker tutorial*

■ **85. GCP / Azure (overview)**	· GCP — Vertex AI, GCS, Cloud Run · Azure — Azure ML, Blob Storage · Cloud Run — serverless containers · Managed endpoints · Cost management basics	*→ Vertex AI* *→ Cloud Run* *→ Azure ML*	■ ***Search:*** *Google Cloud Vertex AI machine learning deployment*
■ **86. CI/CD for ML**	· GitHub Actions — workflows · Automated testing on push · Docker build + push in CI · Model validation in pipeline · Deploy on merge to main · Secrets management · Matrix testing	*→ GitHub Actions* *→ workflow.yml* *→ CI/CD pipeline*	■ ***Search:*** *GitHub Actions CI CD machine learning deployment*

■ Experiment Tracking & MLOps Tools

TOPIC	SUBTOPICS / WHAT TO LEARN	ALGORITHMS / CONCEPTS	SEARCH THIS ON YT / DOCS
■ **87. MLflow**	· Tracking — log params, metrics, artifacts · mlflow.start_run() · MLflow UI — compare experiments · Model registry — staging, production · mlflow.log_model() · MLflow projects · Autolog for sklearn/PyTorch	*→ mlflow.log_metric* *→ mlflow.log_model* *→ Model Registry*	■ ***Search:*** *MLflow experiment tracking tutorial Python*
■ **88. Weights & Biases (W&B;)**	· wandb.init(), wandb.log() · Runs, projects, artifacts · Sweeps — hyperparameter search · W&B; Tables for data visualization · Model versioning · Integration with PyTorch/HuggingFace	*→ wandb.init* *→ wandb.log* *→ Sweeps* *→ Artifacts*	■ ***Search:*** *Weights Biases tutorial ML experiment tracking*
■ **89. Data Versioning & DVC**	· DVC — Data Version Control · dvc add, push, pull · DVC pipelines · Remote storage with S3/GCS · Data and model versioning · Reproducing experiments	*→ dvc add* *→ dvc push* *→ dvc repro*	■ ***Search:*** *DVC data version control tutorial machine learning*
■ **90. Monitoring in Production**	· Data drift — input distribution changes · Concept drift — relationship changes · Model performance monitoring · Evidently AI — drift reports · Prometheus + Grafana basics · Logging — structured logs, ELK stack · Alerting on degradation	*→ Evidently AI* *→ data drift* *→ Prometheus* *→ Grafana*	■ ***Search:*** *ML model monitoring data drift production*

■ Databases & Backend for ML

TOPIC	SUBTOPICS / WHAT TO LEARN	ALGORITHMS / CONCEPTS	SEARCH THIS ON YT / DOCS
✓ **91. SQL & PostgreSQL**	· SELECT, WHERE, ORDER BY, LIMIT · JOINs — INNER, LEFT, RIGHT, FULL · Aggregations — GROUP BY, HAVING · Window functions — ROW_NUMBER, LAG, LEAD · CTEs — WITH clause · Indexes — B-tree, hash, GIN, GIST · Transactions — ACID · Explain, Analyze for query optimization	*→ JOIN* *→ Window functions* *→ CTEs* *→ Indexing* *→ EXPLAIN ANALYZE*	■ ***Search:*** *PostgreSQL advanced SQL tutorial window functions*

✓ 92. SQLAlchemy & Alembic	· ORM models — declarative base · Session, query, filter · Relationships — one-to-many, many-to-many · Alembic migrations — upgrade, downgrade · Async SQLAlchemy · Connection pooling	→ *declarative_base* → *Session* → *alembic upgrade head*	■ ***Search:*** *SQLAlchemy Alembic tutorial Python ORM*
✓ 93. Redis & Celery	· Redis — key-value store, pub/sub · Redis data types — string, hash, list, set · Celery — distributed task queue · Celery workers and beat scheduler · Task states — PENDING, SUCCESS, FAILURE · Retry logic, rate limiting · Flower — Celery monitoring	→ *redis-py* → *celery.task* → *Celery beat* → *Flower*	■ ***Search:*** *Celery Redis async task queue Python tutorial*

PHASE 6

DSA, System Design & Interviews

Weeks 31–36 · Get the job

■ Data Structures & Algorithms

TOPIC	SUBTOPICS / WHAT TO LEARN	ALGORITHMS / CONCEPTS	SEARCH THIS ON YT / DOCS
■ 94. Arrays & Strings	· Two pointers technique · Sliding window · Prefix sum, suffix sum · Kadane's algorithm (max subarray) · String reversal, palindromes · Anagram detection · Matrix traversal — spiral, diagonal	*→ Two pointers* *→ Sliding window* *→ Kadane's* *→ Prefix sum*	■ ***Search:*** *arrays strings interview problems Python LeetCode*
■ 95. Hash Maps & Sets	· HashMap — O(1) lookup · Frequency counting · Two sum, group anagrams · Subarray with given sum · LRU Cache implementation · Consistent hashing	*→ HashMap* *→ Frequency count* *→ LRU Cache*	■ ***Search:*** *hashmap problems Python LeetCode medium*
■ 96. Linked Lists	· Singly, doubly linked list · Reverse linked list · Detect cycle — Floyd's algorithm · Merge sorted lists · Find middle node · Remove nth from end	*→ Floyd's cycle* *→ Slow/fast pointer*	■ ***Search:*** *linked list problems Python interview*
■ 97. Stacks & Queues	· Stack — push, pop, peek · Valid parentheses · Monotonic stack · Queue — FIFO, deque · BFS using queue · Priority Queue / Min-Heap	*→ Monotonic stack* *→ Min-heap* *→ deque*	■ ***Search:*** *stack queue problems Python interview*
■ 98. Trees & Graphs	· Binary tree — inorder, preorder, postorder · Level order traversal (BFS) · DFS — recursive and iterative · Binary Search Tree operations · Lowest Common Ancestor · Graph — adjacency list/matrix · BFS and DFS on graphs · Topological sort · Dijkstra's shortest path · Union-Find (Disjoint Set)	*→ DFS* *→ BFS* *→ Dijkstra* *→ Topological sort* *→ Union-Find*	■ ***Search:*** *trees graphs interview Python LeetCode DFS BFS*
■ 99. Sorting & Searching	· Bubble, selection, insertion sort · Merge sort — O(n log n) · Quick sort — pivot, partitioning · Heap sort · Binary search — standard and variants · Search in rotated array · Find peak element	*→ Merge sort* *→ Quick sort* *→ Binary search variants*	■ ***Search:*** *sorting searching algorithms Python interview*

■ **100. Dynamic Programming**	· Memoization vs tabulation · 1D DP — climbing stairs, house robber · 2D DP — grid paths, edit distance · Knapsack — 0/1, unbounded · Longest Common Subsequence (LCS) · Longest Increasing Subsequence (LIS) · Coin change · DP on strings, intervals, trees	*→ Memoization* *→ Tabulation* *→ Knapsack* *→ LCS* *→ LIS*	■ ***Search:*** *dynamic programming problems Python interview*
■ **101. SQL for ML Interviews**	· Window functions — RANK, DENSE_RANK · LAG, LEAD for time-series queries · Pivot with CASE WHEN · Self joins · Recursive CTEs · Query optimization — indexes · Finding duplicates, nth highest	*→ RANK() OVER* *→ LAG LEAD* *→ Recursive CTE* *→ Self join*	■ ***Search:*** *SQL interview questions window functions advanced*

■ ML System Design

TOPIC	SUBTOPICS / WHAT TO LEARN	ALGORITHMS / CONCEPTS	SEARCH THIS ON YT / DOCS
■ **102. Recommendation System Design**	· Content-based filtering · Collaborative filtering — user-user, item-item · Matrix factorization — SVD, ALS · Two-tower neural model · Cold start problem · Offline vs online evaluation · A/B testing for recommendations	*→ Matrix factorization* *→ Two-tower* *→ Cold start* *→ A/B testing*	■ ***Search:*** *recommendation system design machine learning*
✓ **103. Search System Design**	· Inverted index · TF-IDF ranking · BM25 ranking · Neural search — bi-encoder · Hybrid search — sparse + dense · Query understanding · Elasticsearch basics	*→ Inverted index* *→ BM25* *→ Bi-encoder* *→ Elasticsearch*	■ ***Search:*** *search system design neural search machine learning*
■ **104. ML Platform Design**	· Feature store — offline + online · Training pipeline design · Model registry · A/B testing infrastructure · Shadow mode deployment · Canary releases · Data flywheel	*→ Feature store* *→ Shadow mode* *→ Canary* *→ Data flywheel*	■ ***Search:*** *ML platform system design feature store*
■ **105. Scalability Patterns**	· Horizontal vs vertical scaling · Load balancing · Caching strategies — CDN, Redis · Message queues — Kafka, SQS · Microservices vs monolith · Database sharding, replication · CAP theorem · Rate limiting	*→ Load balancer* *→ Kafka* *→ Sharding* *→ CAP theorem*	■ ***Search:*** *system design scalability patterns software engineer*

■ Interview Preparation

TOPIC	SUBTOPICS / WHAT TO LEARN	ALGORITHMS / CONCEPTS	SEARCH THIS ON YT / DOCS

■ **106. ML Concepts Interview**	· Explain any algorithm from scratch · Bias-variance tradeoff — always asked · How would you handle class imbalance? · What is gradient descent? · Why use log loss for classification? · Difference: parametric vs non-parametric · L1 vs L2 regularization difference · How does XGBoost differ from RF?		■ ***Search:*** *ML interview questions answers explained*
■ **107. Coding Interview**	· LeetCode — solve 75 Blind 75 problems · Practice on Python specifically · Know time/space complexity for each solution · Mock interviews — Pramp, interviewing.io · STAR format for behavioural · Explain your thought process aloud		■ ***Search:*** *LeetCode Blind 75 Python solutions explained*
■ **108. Project Walkthroughs**	· Be able to explain every line of your project · What would you improve if you had more time? · What were the challenges? · How did you evaluate the model? · How does it scale to 10x data? · Explain the architecture diagram		■ ***Search:*** *how to explain ML projects in interviews*